ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 24 cm	Image forms at C = 2f.
2	**(d)** 0.5	0.75/1.5 = 0.5.
3	**(d)** 40 cm	1/v = 1/20 - 1/40 = 1/40.
4	**(a)** 30 cm	Same size at C.
5	**(b)** 1.5×10^8 m/s	v = c/n.
6	**(c)** 3 cm (inverted)	v = 20 cm, m = -1.
7	**(b)** 36 cm	1/v = -1/12 + 1/18 = -1/36.
8	**(a)** $\sqrt{2}$	sin45/sin30 = $(1/\sqrt{2})/(1/2) = \sqrt{2}$.
9	**(c)** 20 cm	1/u = 1/v - 1/f = 1/60 - 1/15 = -1/20.
10	**(a)** 12 cm behind the mirror	1/v = 1/20 + 1/30 = 1/12.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1/f = 1/v + 1/u = -1/30 - 1/30 (1); f = -15 cm (1).	1+1
12	v = c/n = 3×10^8 / 1.6 (1); v = 1.875×10^8 m/s (1).	1+1
13	It always diverges rays (1); Refracted rays appear to meet only when extended backwards (1).	1+1
14	m = -v/u = -(7.5)/(-15) (1); m = +0.5 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	1/v = -1/15 + 1/20 = -1/60 (1); v = -60 cm; m = -v/u = -3 (1); Size 18 cm; real, inverted, magnified (1).	1+1+1
16	1/v = -1/10 - 1/20 = -3/20 (1); v = -6.67 cm (1); m = v/u = 1/3; virtual, erect, diminished (1).	1+1+1
17	Both give real, inverted, same-sized images (1); Mirror image forms on the same side of the object (1); Lens image forms on the opposite side (1).	1+1+1
	OR: 1/v = 1/20 + 1/20 = 1/10 (1); v = +10 cm (1); m = -v/u = 0.5; virtual, erect, diminished (1).	1+1+1
18	1/v = 1/8 - 1/12 = 1/24 (1); v = 24 cm, m = -2 (1); Height 8 cm; real, inverted (1).	1+1+1
	OR: Focal length increases (1); Relative refractive index between lens and water decreases (1); So the lens bends light less (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	1/v = -1/12 + 1/20 = -1/30 (1); v = -30 cm (1); m = -v/u = -1.5 (1); Image size = -4.5 cm (1); Real, inverted, magnified; rays: parallel and through F (1).	1+1+1+1+1
	OR: (a) Water, kerosene, glass, diamond (1); (b) v = 3×10^8/2.42 (1); v = 1.24×10^8 m/s (1); (c) Diamond (1); Greater n means greater bending towards the normal (1).	1+1+1+1+1
20	f = +20 cm (1); 1/v = 1/20 + 1/20 = 1/10, v = +10 cm (1); m = -v/u = +0.5 (1); Height 2.5 cm (1); Virtual, erect, diminished behind the mirror (1).	1+1+1+1+1
	OR: Convex: v = 30 cm, m = -2, real inverted magnified (1); 1/v = 1/10 - 1/15 = 1/30 (1); Concave: 1/v = -1/10 - 1/15 = -1/6 (1); v = -6 cm (1); m = 0.4, virtual erect diminished (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.

- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** water	Water has a lower n.
2	**(c)** virtual, erect and three times larger	Positive m means virtual and erect.
3	**(a)** +2	v = +20 cm, m = -v/u = 2.
4	**(c)** along the normal	Normal incidence does not bend.
5	**(d)** 15 cm on the same side	1/v = -1/30 - 1/30 = -1/15.
6	**(a)** plane	m = +1.
7	**(c)** Thicker slab	Displacement grows with thickness.
8	**(c)** +2 D	P = P1 + P2.
9	**(a)** concave, with the object between P and F	Only concave can magnify with virtual image.
10	**(d)** the speed of light in it is lower	n depends on speed.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	sin r = sin i / n = 0.5/1.5 (1); sin r = 1/3 (1).	1+1
12	f = 1/4 m = 25 cm (1); Positive: converging (convex) (1).	1+1
13	Object must be placed between pole and focus (1); Object distance less than 10 cm (1).	1+1
14	f = 1/P = -0.4 m = -40 cm (1); Concave lens (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	m = -3 = -v/u, v = 3u = -30 cm (1); 1/f = 1/v + 1/u = -1/30 - 1/10 (1); f = -7.5 cm (1).	1+1+1
16	P1 = +4 D (1); P2 = -2.5 D (1); Net P = +1.5 D (1).	1+1+1
17	1/v = -1/20 + 1/25 = -1/100 (1); v = -100 cm (1); Real, inverted, magnified (m = -4) (1).	1+1+1
	OR: sin r = sin 60 / 1.5 = 0.866/1.5 = 0.577 (1); v = c/n (1); v = 2×10^8 m/s (1).	1+1+1
18	1/v = 1/12 - 1/8 = -1/24 (1); v = -24 cm, m = v/u = 3 (1); Virtual, erect, magnified (1).	1+1+1
	OR: f = -25 cm (1); 1/v = -1/25 - 1/50 = -3/50 (1); v = -16.7 cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	u = -30, v = +60 (1); 1/f = 1/60 - 1/(-30) = 1/60 + 1/30 = 1/20 (1); f = +20 cm convex lens (1); m = v/u = -2 (1); Image size 4 cm, real, inverted (1).	1+1+1+1+1
	OR: P = +5 - 3 = +2 D (1); f = 50 cm (1); 1/v = 1/50 - 1/100 = 1/100 (1); v = +100 cm (1); Real image at 100 cm on the other side (1).	1+1+1+1+1
20	sin r = sin 30 / $\sqrt{3}$ = 1/(2 $\sqrt{3}$) (1); r = about 16.8 degrees (sin r = 0.289) (1); At the second face the ray bends away (1); Faces are parallel, so emergent angle equals incident angle (1); Angle of emergence = 30 degrees (1).	1+1+1+1+1
	OR: Real: rays actually meet, can be obtained on a screen (1); Virtual: rays appear to meet, cannot be obtained on a screen (1); Real: object beyond F (projector or solar furnace) (1); Virtual: object between P and F (1); Application: shaving mirror, dentist's mirror (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.

- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Convex with object between F and O	Magnifier condition.
2	**(a)** real inverted image twice as big	Negative sign indicates real and inverted.
3	**(a)** 2/3	Reciprocal.
4	**(b)** 10 cm	Object at 2F: 2f = 20 cm.
5	**(d)** infinity	Object is at F.
6	**(b)** Nearer to the surface than the actual position	Refraction raises the apparent depth.
7	**(d)** +4 D	P = 1/0.25.
8	**(b)** Image on a screen from a concave mirror	Only the first can be received on a screen.
9	**(d)** Angle of emergence = angle of incidence	Parallel faces.
10	**(a)** at F, virtual, point-sized	Image forms at the focus on the same side.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1/v = 1/5 - 1/10 = 1/10, v = 10 cm (1); m = -1, size = 2 cm inverted (1).	1+1
12	Speed of light decreases in the denser medium (1); Wavefront turns towards the slower side (1).	1+1
13	1/v = 1/10 - 1/15 = 1/30 (1); v = 30 cm (1).	1+1
14	Rays bend away from normal while emerging (1); Rays appear to originate at a shallower depth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	n(g,w) = n_g / n_w (1); 1.5/1.33 (1); = 1.13 (1).	1+1+1
16	u is between 0 and f (1); 1/v = 1/f + 1/u becomes negative (1); v negative and |v| > |u|, so magnified and virtual (1).	1+1+1
17	The ray diverges after reflection (1); It appears to come from the principal focus behind (1); The mirror is diverging (1).	1+1+1
	OR: 1/u = 1/v - 1/f = 1/60 - 1/20 = -1/30 (1); u = -30 cm (1); m = v/u = -2 (1).	1+1+1
18	v = +30, u = -10 (1); 1/f = 1/30 - 1/10 = -1/15 (1); f = -15 cm (1).	1+1+1
	OR: If v is negative, the image is real (in front) (1); If v is positive, the image is virtual (behind) (1); Sign of m gives erect or inverted (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Object between O and F (less than 10 cm) (1); 1/v = 1/10 - 1/6 = -1/15 (1); v = -15 cm (1); m = v/u = 2.5 (1); Virtual, erect, magnified (1).	1+1+1+1+1
	OR: Object is placed at C (1); f = R/2 = -18 cm (1); u = -2f = -36 cm (1); Image forms at C, real, inverted (1); m = -1 (1).	1+1+1+1+1
20	m = 1/3 = v/u, v = -10 cm (1); 1/f = 1/v - 1/u = -1/10 + 1/30 (1); f = -15 cm (1); P = 1/(-0.15) = -6.67 D (1); Concave lens (1).	1+1+1+1+1
	OR: 1/v - 1/u = 1/f (1); m = v/u (1); 1/v = 1/f + 1/u = 1/f - 1/(2f) = 1/(2f) (1); v = 2f, so m = -1 (1); Real, inverted, same-sized image (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.

- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND